

Simple Harmonic Motion — Solutions

Warm-up

Solution to Question 1:

Given the position $x = \cos(t)$. To prove Simple Harmonic Motion (SHM), we must find the second derivative of x with respect to time t and show it is proportional to the displacement from a center c .

$$\begin{aligned}\dot{x} &= \frac{d}{dt}(\cos(t)) \\ &= -\sin(t) \\ \ddot{x} &= \frac{d}{dt}(-\sin(t)) \\ &= -\cos(t) \\ \ddot{x} &= -(x) \\ \ddot{x} &= -1^2(x - 0)\end{aligned}$$

Thus, the particle is undergoing SHM with $n = 1$ and $c = 0$.

Solution to Question 2a:

Given $x = 3\sin(t) + 2$.

$$\begin{aligned}\dot{x} &= 3\cos(t) \\ \ddot{x} &= -3\sin(t) \\ \ddot{x} &= -(3\sin(t)) \\ \ddot{x} &= -(x - 2) \\ \ddot{x} &= -1^2(x - 2)\end{aligned}$$

This matches the form $\ddot{x} = -n^2(x - c)$ with $n = 1$ and $c = 2$.

Solution to Question 2b:

Given $\dot{x} = 3\cos(t)$. First, we find the acceleration $\ddot{x}$.

$$\begin{aligned}\ddot{x} &= \frac{d}{dt}(3\cos(t)) \\ &= -3\sin(t)\end{aligned}$$

Now we need to express $\ddot{x}$ in terms of x . Integrating $\dot{x}$:

$$\begin{aligned}\int dx &= \int 3\cos(t)dt \\ x &= 3\sin(t) + c\end{aligned}$$

Substituting $3\sin(t) = x - c$ into the acceleration equation:

$$\ddot{x} = -(x - c)$$

$$\ddot{x} = -1^2(x - c)$$

This matches the form $\ddot{x} = -n^2(x - c)$ with $n = 1$ and c being the constant of integration.

Solution to Question 2c:

Given $x = 4 \sin^2(t) + 5$. We use the identity $\sin^2(t) = \frac{1 - \cos(2t)}{2}$.

$$x = 4 \left(\frac{1 - \cos(2t)}{2} \right) + 5$$

$$x = 2(1 - \cos(2t)) + 5$$

$$x = 2 - 2 \cos(2t) + 5$$

$$x = -2 \cos(2t) + 7$$

Now find the derivatives:

$$\dot{x} = -2(-\sin(2t) \cdot 2)$$

$$= 4 \sin(2t)$$

$$\ddot{x} = 4(\cos(2t) \cdot 2)$$

$$= 8 \cos(2t)$$

From $x = -2 \cos(2t) + 7$, we have $2 \cos(2t) = 7 - x$, so $8 \cos(2t) = 4(7 - x)$.

$$\ddot{x} = 4(7 - x)$$

$$\ddot{x} = -4(x - 7)$$

$$\ddot{x} = -2^2(x - 7)$$

This matches the form $\ddot{x} = -n^2(x - c)$ with $n = 2$ and $c = 7$.

Solution to Question 3a:

Given $x = 4 \sin(t) + 3$. The general form is $x = A \sin(nt + \theta) + c$.

$$\text{Amplitude } A = 4$$

Solution to Question 3b:

Given $x = 4 \sin(t) + 3$.

$$\text{Centre } c = 3$$

Solution to Question 3c:

The extremities are $c \pm A$.

$$\begin{aligned}\text{Extremities} &= 3 \pm 4 \\ \text{Lower bound} &= 3 - 4 = -1 \\ \text{Upper bound} &= 3 + 4 = 7\end{aligned}$$

Solution to Question 3d:

The period T is given by $T = \frac{2\pi}{n}$. Here $n = 1$.

$$T = \frac{2\pi}{1} = 2\pi \text{ seconds}$$

Solution to Question 3e:

Set $x = 1$ in the equation $x = 4 \sin(t) + 3$.

$$\begin{aligned}1 &= 4 \sin(t) + 3 \\ -2 &= 4 \sin(t) \\ \sin(t) &= -\frac{1}{2}\end{aligned}$$

For the first time $t > 0$, t must be in the third quadrant:

$$\begin{aligned}t &= \pi + \frac{\pi}{6} \\ t &= \frac{7\pi}{6} \text{ seconds}\end{aligned}$$

Solution to Question 3f:

First, find the velocity expression $\dot{x}$.

$$\begin{aligned}\dot{x} &= \frac{d}{dt}(4 \sin(t) + 3) \\ &= 4 \cos(t)\end{aligned}$$

Substitute $t = \frac{7\pi}{6}$:

$$\begin{aligned}\dot{x} &= 4 \cos\left(\frac{7\pi}{6}\right) \\ \dot{x} &= 4 \left(-\frac{\sqrt{3}}{2}\right) \\ \dot{x} &= -2\sqrt{3} \text{ m/s}\end{aligned}$$

Solution to Question 4a:

Given: Period $T = 4\pi$, Centre $c = 2$, Amplitude $A = 5$.

$$T = \frac{2\pi}{n} \implies 4\pi = \frac{2\pi}{n} \implies n = \frac{1}{2}$$

Substituting $n = \frac{1}{2}$ and $c = 2$ into $\ddot{x} = -n^2(x - c)$:

$$\begin{aligned}\ddot{x} &= -\left(\frac{1}{2}\right)^2 (x - 2) \\ \ddot{x} &= -\frac{1}{4}(x - 2)\end{aligned}$$

Solution to Question 4b:

The general velocity equation for SHM is $v^2 = n^2(A^2 - (x - c)^2)$. Expanding this:

$$\begin{aligned}v^2 &= n^2(A^2 - (x^2 - 2cx + c^2)) \\ v^2 &= n^2(-x^2 + 2cx + A^2 - c^2)\end{aligned}$$

Substitute $n = \frac{1}{2}$, $A = 5$, $c = 2$:

$$\begin{aligned}v^2 &= \left(\frac{1}{2}\right)^2 (-x^2 + 2(2)x + 5^2 - 2^2) \\ v^2 &= \frac{1}{4}(-x^2 + 4x + 21)\end{aligned}$$

Solution to Question 4c:

Using the form $v^2 = n^2(A^2 - (x - c)^2)$ with $n = \frac{1}{2}$, $A = 5$, $c = 2$:

$$\begin{aligned}v^2 &= \left(\frac{1}{2}\right)^2 (5^2 - (x - 2)^2) \\ v^2 &= \frac{1}{4}(25 - (x - 2)^2)\end{aligned}$$

Solution to Question 4d:

General form: $x = A \sin(nt + \theta) + c$. Given $A = 5$, $n = 1/2$, $c = 2$. At $t = 0$, $x = 2$ and moving in positive direction ($\dot{x} > 0$).

$$\begin{aligned}2 &= 5 \sin(1/2(0) + \theta) + 2 \\ 0 &= 5 \sin(\theta) \\ \sin(\theta) &= 0 \implies \theta = 0, \pi, \dots\end{aligned}$$

Check velocity $\dot{x} = 5(1/2) \cos(1/2t + \theta)$. At $t = 0$, $\dot{x} = 2.5 \cos(\theta)$. For $\dot{x} > 0$, $\theta = 0$.

$$x = 5 \sin\left(\frac{1}{2}t\right) + 2$$

Solution to Question 5a:

Given $x = 4 \sin(t) + 3 \cos(t) + 3$. We use the harmonic addition theorem $a \sin(t) + b \cos(t) = R \sin(t + \alpha)$.

$$R = \sqrt{4^2 + 3^2} = \sqrt{16 + 9} = 5$$

Thus, the amplitude $A = 5$.

Solution to Question 5b:

The center is $c = 3$. The extremities are $c \pm A$.

$$\text{Extremities} = 3 \pm 5$$

$$\text{Lower bound} = 3 - 5 = -2$$

$$\text{Upper bound} = 3 + 5 = 8$$

Solution to Question 5c:

The coefficient of t is $n = 1$.

$$T = \frac{2\pi}{1} = 2\pi \text{ seconds}$$

Solution to Question 5d:

First, express x as a single sine function. $x = 5 \sin(t + \alpha) + 3$ where $\tan \alpha = 3/4$.

$$\alpha = \tan^{-1}(0.75) \approx 0.6435 \text{ rad}$$

Set $x = 1$:

$$1 = 5 \sin(t + 0.6435) + 3$$

$$-2 = 5 \sin(t + 0.6435)$$

$$\sin(t + 0.6435) = -0.4$$

Find the first positive t :

$$t + 0.6435 = \pi + \sin^{-1}(0.4)$$

$$t + 0.6435 = 3.1416 + 0.4115$$

$$t = 3.5531 - 0.6435$$

$$t \approx 2.91 \text{ seconds}$$

Solution to Question 5e:

Find velocity $\dot{x} = 4 \cos(t) - 3 \sin(t)$.

$$\begin{aligned}\dot{x} &= 4 \cos(2.91) - 3 \sin(2.91) \\ \dot{x} &\approx 4(-0.972) - 3(0.230) \\ \dot{x} &\approx -3.888 - 0.690 \\ \dot{x} &\approx -4.58 \text{ m/s}\end{aligned}$$

Solution to Question 5f:

The graph is a sine wave shifted up by 3 units. - Equilibrium position: $x = 3$ - Amplitude: 5 - Extremities: $x = -2$ and $x = 8$ - Period: $2\pi \approx 6.28$

Skill-building

Solution to Question 1a:

Given $x = 3 \sin(t) + 2 \cos(t) + 2$.

$$\begin{aligned}\dot{x} &= 3 \cos(t) - 2 \sin(t) \\ \ddot{x} &= -3 \sin(t) - 2 \cos(t) \\ \ddot{x} &= -(3 \sin(t) + 2 \cos(t)) \\ \ddot{x} &= -(x - 2) \\ \ddot{x} &= -1^2(x - 2)\end{aligned}$$

This is SHM with $n = 1$ and $c = 2$.

Solution to Question 1b:

Given $\dot{x} = 3 \cos(t) + \sin(t)$.

$$\ddot{x} = -3 \sin(t) + \cos(t)$$

Integrate $\dot{x}$ to find x :

$$\begin{aligned}x &= \int (3 \cos(t) + \sin(t)) dt \\ x &= 3 \sin(t) - \cos(t) + c\end{aligned}$$

Observe that $3 \sin(t) - \cos(t) = x - c$. Now look at $\ddot{x}$:

$$\begin{aligned}\ddot{x} &= -(3 \sin(t) - \cos(t)) \\ \ddot{x} &= -(x - c)\end{aligned}$$

This is SHM with $n = 1$ and c as the constant of integration.

Solution to Question 1c:

Given $x = 4 \sin^2(t) + 2 \cos^2(t) + 5$.

$$\begin{aligned}
x &= 2(2\sin^2(t) + \cos^2(t)) + 5 \\
x &= 2(\sin^2(t) + 1) + 5 \\
x &= 2\sin^2(t) + 7
\end{aligned}$$

Using $\sin^2(t) = \frac{1-\cos(2t)}{2}$:

$$\begin{aligned}
x &= 2\left(\frac{1-\cos(2t)}{2}\right) + 7 \\
x &= 1 - \cos(2t) + 7 \\
x &= -\cos(2t) + 8
\end{aligned}$$

Derivatives:

$$\begin{aligned}
\dot{x} &= 2\sin(2t) \\
\ddot{x} &= 4\cos(2t)
\end{aligned}$$

From $x = -\cos(2t) + 8$, we have $\cos(2t) = 8 - x$.

$$\begin{aligned}
\ddot{x} &= 4(8 - x) \\
\ddot{x} &= -4(x - 8) \\
\ddot{x} &= -2^2(x - 8)
\end{aligned}$$

This is SHM with $n = 2$ and $c = 8$.

Solution to Question 2a:

Given $x = 6\sin^2(t) + 7$.

$$\begin{aligned}
x &= 6\left(\frac{1-\cos(2t)}{2}\right) + 7 \\
x &= 3(1 - \cos(2t)) + 7 \\
x &= 3 - 3\cos(2t) + 7 \\
x &= -3\cos(2t) + 10
\end{aligned}$$

Comparing to $x = A\cos(nt + \theta) + c$, we find $A = 3$ (amplitude is always positive).

Solution to Question 2b:

From the simplified form $x = -3\cos(2t) + 10$, the center is the constant term.

$$c = 10$$

Solution to Question 2c:

Extremities are $c \pm A$.

$$\begin{aligned}\text{Extremities} &= 10 \pm 3 \\ \text{Lower bound} &= 7 \\ \text{Upper bound} &= 13\end{aligned}$$

Solution to Question 2d:
The coefficient of t is $n = 2$.

$$T = \frac{2\pi}{2} = \pi \text{ seconds}$$

Solution to Question 2e:
Set $x = 10$ in $x = -3 \cos(2t) + 10$.

$$\begin{aligned}10 &= -3 \cos(2t) + 10 \\ 0 &= -3 \cos(2t) \\ \cos(2t) &= 0\end{aligned}$$

For the first positive t :

$$\begin{aligned}2t &= \frac{\pi}{2} \\ t &= \frac{\pi}{4} \text{ seconds}\end{aligned}$$

Solution to Question 2f:
Find velocity $\dot{x} = 6 \sin(2t)$.

$$\begin{aligned}\dot{x} &= 6 \sin\left(2 \cdot \frac{\pi}{4}\right) \\ \dot{x} &= 6 \sin\left(\frac{\pi}{2}\right) \\ \dot{x} &= 6(1) = 6 \text{ m/s}\end{aligned}$$

Solution to Question 3a:
Given: Period $T = 2$, Centre $c = 1$, Amplitude $A = 3$.

$$T = \frac{2\pi}{n} \implies 2 = \frac{2\pi}{n} \implies n = \pi$$

Substituting into $\ddot{x} = -n^2(x - c)$:

$$\ddot{x} = -\pi^2(x - 1)$$

Solution to Question 3b:

Using the form $v^2 = n^2(A^2 - (x - c)^2)$ with $n = \pi, A = 3, c = 1$:

$$\begin{aligned}v^2 &= \pi^2(3^2 - (x - 1)^2) \\v^2 &= \pi^2(9 - (x - 1)^2)\end{aligned}$$

Here, n is the angular frequency, A is the amplitude, and c is the centre of motion.

Solution to Question 3c:

General form: $x = A \sin(nt + \theta) + c$. Given $A = 3, n = \pi, c = 1$. At $t = 0$, the particle is at the furthest point from the origin. Since $c = 1$ and $A = 3$, the furthest point is $1 + 3 = 4$ (or $1 - 3 = -2$). The problem states it moves back towards the origin, so it starts at $x = 4$.

$$\begin{aligned}4 &= 3 \sin(\pi(0) + \theta) + 1 \\3 &= 3 \sin(\theta) \\\sin(\theta) &= 1 \implies \theta = \frac{\pi}{2}\end{aligned}$$

Thus:

$$x = 3 \sin\left(\pi t + \frac{\pi}{2}\right) + 1$$

Solution to Question 4a:

Given $x = 4 \sin(t) + 3 \cos(t) + 3$. This is identical to Warm-up Question 5.

$$R = \sqrt{4^2 + 3^2} = 5$$

Amplitude $A = 5$.

Solution to Question 4b:

Centre $c = 3$. Extremities are 3 ± 5 .

$$\text{Extremities} = -2, 8$$

Solution to Question 4c:

Coefficient of t is $n = 1$.

$$T = \frac{2\pi}{1} = 2\pi \text{ seconds}$$

Solution to Question 4d:

Identical to Warm-up 5d.

$$t \approx 2.91 \text{ seconds}$$

Solution to Question 4e:

Identical to Warm-up 5e.

$$\dot{x} \approx -4.58 \text{ m/s}$$

Solution to Question 4f:

Identical to Warm-up 5f.

Easier Exam Questions

Solution to Question 1i:

Given $v^2 = 12 + 4x - x^2$. To show SHM, we differentiate with respect to t .

$$\begin{aligned}\frac{d}{dt}(v^2) &= \frac{d}{dt}(12 + 4x - x^2) \\ 2v\dot{v} &= (4 - 2x)\dot{x} \\ 2v\ddot{x} &= (4 - 2x)v \\ \ddot{x} &= 2 - x \\ \ddot{x} &= -1(x - 2)\end{aligned}$$

This is in the form $\ddot{x} = -n^2(x - c)$ with $n = 1$ and $c = 2$. The centre of motion is $x = 2$.

Solution to Question 1ii:

From $\ddot{x} = -1(x - 2)$, we have $n = 1$.

$$T = \frac{2\pi}{n} = \frac{2\pi}{1} = 2\pi \text{ seconds}$$

To find the amplitude A , we complete the square for $v^2 = 0$ at the extremities:

$$\begin{aligned}0 &= -x^2 + 4x + 12 \\ 0 &= -(x^2 - 4x - 12) \\ 0 &= -(x - 6)(x + 2) \\ x &= 6, -2\end{aligned}$$

Amplitude $A = \frac{6 - (-2)}{2} = 4$ metres.

Solution to Question 1iii:

General form: $x = a \sin(nt + \theta) + b$. We know $n = 1, b = c = 2, a = A = 4$.

$$x = 4 \sin(t + \theta) + 2$$

Initially, the particle is 6 metres to the right of the origin ($t = 0, x = 6$):

$$6 = 4 \sin(\theta) + 2$$

$$4 = 4 \sin(\theta)$$

$$\sin(\theta) = 1 \implies \theta = \frac{\pi}{2}$$

Thus, $\theta = \frac{\pi}{2}$ and $b = 2$.

Solution to Question 2i:

Given $v^2 = 189 - 42x - 7x^2$. Differentiate with respect to t :

$$2v\dot{v} = (-42 - 14x)\dot{x}$$

$$2v\ddot{x} = (-42 - 14x)v$$

$$\ddot{x} = -21 - 7x$$

$$\ddot{x} = -7(x + 3)$$

This is in the form $\ddot{x} = -n^2(x - c)$ with $n^2 = 7$ and $c = -3$.

Solution to Question 2ii:

From $\ddot{x} = -7(x + 3)$, we have $n = \sqrt{7}$.

$$T = \frac{2\pi}{\sqrt{7}} = \frac{2\pi\sqrt{7}}{7} \text{ seconds}$$

Solution to Question 2iii:

Set $v^2 = 0$ to find extremities:

$$0 = -7x^2 - 42x + 189$$

$$0 = x^2 + 6x - 27$$

$$0 = (x + 9)(x - 3)$$

$$x = -9, 3$$

Amplitude $A = \frac{3 - (-9)}{2} = 6$ metres.

Solution to Question 2iv:

Maximum speed occurs at the centre of motion $x = c = -3$.

$$v_{max}^2 = 189 - 42(-3) - 7(-3)^2$$

$$v_{max}^2 = 189 + 126 - 63$$

$$v_{max}^2 = 252$$

$$v_{max} = \sqrt{252} = 6\sqrt{7} \text{ m/s}$$

Solution to Question 3i:

Given $\ddot{x} = -4x$. The endpoints occur when $v = 0$. We use the energy relation $v^2 = n^2(A^2 - x^2)$ since $c = 0$.

$$n^2 = 4 \implies n = 2$$

Given $t = 0, x = 0, v = 3$:

$$3^2 = 2^2(A^2 - 0^2)$$

$$9 = 4A^2$$

$$A^2 = \frac{9}{4} \implies A = 1.5$$

Endpoints are $x = \pm 1.5$ metres.

Solution to Question 3ii:

Using $v^2 = 4(1.5^2 - x^2)$:

$$v^2 = 4(2.25 - 1^2)$$

$$v^2 = 4(1.25)$$

$$v^2 = 5$$

$$v = \sqrt{5} \text{ m/s}$$

Solution to Question 4i:

Given $x = 4 \cos(t/2) - 4\sqrt{3} \sin(t/2)$.

$$\dot{x} = 4(-\sin(t/2) \cdot 1/2) - 4\sqrt{3}(\cos(t/2) \cdot 1/2)$$

$$\dot{x} = -2 \sin(t/2) - 2\sqrt{3} \cos(t/2)$$

$$\ddot{x} = -2(\cos(t/2) \cdot 1/2) - 2\sqrt{3}(-\sin(t/2) \cdot 1/2)$$

$$\ddot{x} = -\cos(t/2) + \sqrt{3} \sin(t/2)$$

Observe that $\ddot{x} = -\frac{1}{4}(4 \cos(t/2) - 4\sqrt{3} \sin(t/2)) = -\frac{1}{4}x$.

$$\ddot{x} = -\left(\frac{1}{2}\right)^2 x$$

This is SHM with $n = 1/2$ and $c = 0$.

Solution to Question 4ii:

Express $x = 4 \cos(t/2) - 4\sqrt{3} \sin(t/2)$ in the form $A \cos(t/2 + \alpha)$.

$$A = \sqrt{4^2 + (-4\sqrt{3})^2} = \sqrt{16 + 48} = \sqrt{64} = 8$$

To find α , we use $A \cos(t/2 + \alpha) = A(\cos(t/2) \cos \alpha - \sin(t/2) \sin \alpha)$.

$$\begin{aligned} A \cos \alpha = 4 &\implies 8 \cos \alpha = 4 \implies \cos \alpha = 1/2 \\ A \sin \alpha = 4\sqrt{3} &\implies 8 \sin \alpha = 4\sqrt{3} \implies \sin \alpha = \sqrt{3}/2 \end{aligned}$$

Thus $\alpha = \pi/3$. The equation is $x = 8 \cos(t/2 + \pi/3)$.

$$\begin{aligned} \text{Amplitude } A &= 8 \\ \text{Period } T &= \frac{2\pi}{1/2} = 4\pi \text{ seconds} \end{aligned}$$

Solution to Question 5i:

Given $\ddot{x} = -4(x - 3)$. This is already in the form $\ddot{x} = -n^2(x - c)$ with $n = 2$ and $c = 3$.

$$\text{Oscillation range: } v^2 = n^2(A^2 - (x - c)^2)$$

We are given that at $x = 0$, $v = 8$:

$$\begin{aligned} 8^2 &= 2^2(A^2 - (0 - 3)^2) \\ 64 &= 4(A^2 - 9) \\ 16 &= A^2 - 9 \\ A^2 = 25 &\implies A = 5 \end{aligned}$$

The particle oscillates between $c - A$ and $c + A$:

$$\begin{aligned} \text{Lower bound} &= 3 - 5 = -2 \\ \text{Upper bound} &= 3 + 5 = 8 \end{aligned}$$

It oscillates between $x = -2$ and $x = 8$.

Solution to Question 5ii:

General form: $x = A \cos(nt + \theta) + c$. Given $A = 5, n = 2, c = 3$. At $t = 0, x = 5.5$ and moving towards origin ($\dot{x} < 0$).

$$\begin{aligned} 5.5 &= 5 \cos(2(0) + \theta) + 3 \\ 2.5 &= 5 \cos(\theta) \\ \cos(\theta) = 0.5 &\implies \theta = \pm \frac{\pi}{3} \end{aligned}$$

Since $\dot{x} = -5(2) \sin(2t + \theta)$, at $t = 0$, $\dot{x} = -10 \sin(\theta)$. For $\dot{x} < 0$, $\sin(\theta) > 0$, so $\theta = \pi/3$.

$$x = 5 \cos(2t + \pi/3) + 3$$

Set $x = 0$:

$$\begin{aligned}0 &= 5 \cos(2t + \pi/3) + 3 \\-0.6 &= \cos(2t + \pi/3) \\2t + \pi/3 &= \cos^{-1}(-0.6) \\2t + 1.0472 &= 2.2143 \\2t &= 1.1671 \\t &\approx 0.58 \text{ seconds}\end{aligned}$$

Harder Exam Questions

Solution to Question 1i:

Given $v^2 = 9(12 + 4x - x^2)$. Differentiate with respect to t :

$$\begin{aligned}2v\dot{v} &= 9(4 - 2x)\dot{x} \\2v\ddot{x} &= 9(4 - 2x)v \\\ddot{x} &= \frac{9}{2}(4 - 2x) \\\ddot{x} &= 18 - 9x \\\ddot{x} &= -9(x - 2)\end{aligned}$$

This is in the form $\ddot{x} = -n^2(x - c)$ with $n = 3$ and $c = 2$. Thus, the ball is moving in SHM.

Solution to Question 1ii:

The ball can move where $v^2 \geq 0$.

$$\begin{aligned}9(12 + 4x - x^2) &\geq 0 \\x^2 - 4x - 12 &\leq 0 \\(x - 6)(x + 2) &\leq 0\end{aligned}$$

Thus, the range of x values is $-2 \leq x \leq 6$.

Solution to Question 1iii:

The central position is $c = 2$. The extreme zone is where $|x - 2| > 2$, meaning $x > 4$ or $x < 0$.

$$x = 4 \sin(3t + \theta) + 2$$

Set $x = 4$ to find the boundary of the extreme zone:

$$\begin{aligned}4 &= 4 \sin(3t + \theta) + 2 \\2 &= 4 \sin(3t + \theta) \\\sin(3t + \theta) &= 0.5\end{aligned}$$

This occurs at $3t + \theta = \pi/6$ and $3t + \theta = 5\pi/6$. The time spent in the non-extreme zone (between $x = 0$ and $x = 4$) is the time between these two values:

$$\Delta(3t + \theta) = 5\pi/6 - \pi/6 = 4\pi/6 = 2\pi/3$$

The total period is 2π for the argument $3t + \theta$. The time spent in the non-extreme zone per half-cycle is $2\pi/3$. In one full cycle, the time spent in the non-extreme zone is $2 \cdot (2\pi/3) = 4\pi/3$. The time spent in the extreme zone is $2\pi - 4\pi/3 = 2\pi/3$.

$$\text{Percentage} = \frac{2\pi/3}{2\pi} \times 100 = \frac{1}{3} \times 100 \approx 33\%$$

Solution to Question 2i:

Given $x = 1 + \sqrt{2} \cos(t - \pi/4)$.

$$\dot{x} = -\sqrt{2} \sin(t - \pi/4)$$

$$\ddot{x} = -\sqrt{2} \cos(t - \pi/4)$$

From the original equation, $\sqrt{2} \cos(t - \pi/4) = x - 1$.

$$\ddot{x} = -(x - 1)$$

Solution to Question 2ii:

The particle passes through point O when $x = 0$.

$$0 = 1 + \sqrt{2} \cos(t - \pi/4)$$

$$-1 = \sqrt{2} \cos(t - \pi/4)$$

$$\cos(t - \pi/4) = -\frac{1}{\sqrt{2}}$$

For the first positive t :

$$t - \pi/4 = \frac{3\pi}{4}$$

$$t = \pi \text{ seconds}$$

Solution to Question 2iii:

Average speed is $\frac{\text{Total Distance}}{\text{Period}}$. Total distance in one oscillation is $4A$.

$$A = \sqrt{2}, T = 2\pi/1 = 2\pi$$

$$\text{Average Speed} = \frac{4\sqrt{2}}{2\pi} = \frac{2\sqrt{2}}{\pi} \text{ m/s}$$

Solution to Question 3:

Given $c = 2, A = 3$. The velocity equation is $v^2 = n^2(A^2 - (x - c)^2)$.

$$\begin{aligned} v^2 &= n^2(3^2 - (x - 2)^2) \\ v^2 &= n^2(9 - (x^2 - 4x + 4)) \\ v^2 &= n^2(-x^2 + 4x + 5) \end{aligned}$$

Comparing this to $v^2 = p + qx - 6x^2$:

$$\begin{aligned} -n^2x^2 &= -6x^2 \implies n^2 = 6 \\ q &= 4n^2 = 4(6) = 24 \\ p &= 5n^2 = 5(6) = 30 \end{aligned}$$

Period $T = \frac{2\pi}{\sqrt{6}} = \frac{2\pi\sqrt{6}}{6} = \frac{\pi\sqrt{6}}{3}$ seconds.

Solution to Question 4:

Given $\ddot{x} = -9(x + 2)$. This is SHM with $n = 3$ and $c = -2$.

$$\begin{aligned} v^2 &= n^2(A^2 - (x - c)^2) \\ v^2 &= 9(A^2 - (x + 2)^2) \end{aligned}$$

Given $x = 1, v = 12$:

$$\begin{aligned} 12^2 &= 9(A^2 - (1 + 2)^2) \\ 144 &= 9(A^2 - 9) \\ 16 &= A^2 - 9 \\ A^2 &= 25 \implies A = 5 \end{aligned}$$

Substitute $A = 5$ back into the velocity equation:

$$\begin{aligned} v^2 &= 9(25 - (x^2 + 4x + 4)) \\ v^2 &= 9(21 - 4x - x^2) \\ v^2 &= 189 - 36x - 9x^2 \end{aligned}$$

Period $T = \frac{2\pi}{3}$ seconds, Amplitude $A = 5$ m, Centre $c = -2$ m.

Solution to Question 5i:

Given: High tide $x_{max} = 12$, Low tide $x_{min} = 2$.

$$c = \frac{12+2}{2} = 7$$

$$A = \frac{12-2}{2} = 5$$

Period $T = 2 \times 4 = 8$ hours. Thus $n = \frac{2\pi}{8} = \frac{\pi}{4}$.

At $t = 0$, it is at low tide ($x = 2$):

$$2 = 7 + 5 \cos\left(\frac{\pi}{4}(0) + \theta\right)$$

$$-5 = 5 \cos(\theta)$$

$$\cos(\theta) = -1 \implies \theta = \pi$$

Using $x = c + A \cos(nt + \theta)$:

$$x = 7 + 5 \cos\left(\frac{\pi t}{4} + \pi\right)$$

$$x = 7 - 5 \cos\left(\frac{\pi t}{4}\right) \quad (\text{since } \cos(\theta + \pi) = -\cos \theta)$$

Solution to Question 5ii:

Set $x = 5$ in the equation $x = 7 - 5 \cos(\frac{\pi t}{4})$:

$$5 = 7 - 5 \cos\left(\frac{\pi t}{4}\right)$$

$$-2 = -5 \cos\left(\frac{\pi t}{4}\right)$$

$$\cos\left(\frac{\pi t}{4}\right) = 0.4$$

Find the first positive t :

$$\frac{\pi t}{4} = \cos^{-1}(0.4)$$

$$\frac{\pi t}{4} \approx 1.159$$

$$t = \frac{1.159 \times 4}{\pi}$$

$$t \approx 1.48 \text{ hours}$$

Since t is measured from 2 am, the time is approximately 3:29 am.